

Differentiation Laws CSC1035

Exponential and Log laws

$$(e^x)' = e^x$$

$$(\ln x)' = \frac{1}{x}$$

General rules:

$$(e^g)' = g' * e^g$$

$$(\ln g)' = \frac{g'}{g}$$

Composition Laws

$$(h(g(x)))' = g'(x) * h'(g(x))$$

Division

If $f(x) = g(x) / h(x)$ then:

$$(g/h)' = \frac{g' * h - g * h'}{h^2}$$

Addition and subtraction

If $f(x) = g(x) \pm h(x)$ then:

$$(g \pm h)' = g' \pm h'$$

Multiplication with no constant but two functions.

If $f(x) = g(x) * h(x)$ then:

$$(g * h)' = g' * h + g * h'$$

Constant multiplier

$$(c * g)' = c * g'$$

Constant power

If $f(x) = x^n$ (n can be fractional and/or negative) then:

$$(x^n)' = nx^{n-1}, \text{ if } n \text{ is a constant}$$

E.g. $(x^3)' = 3x^2$ $(\sqrt{x})' = \left(x^{\frac{1}{2}}\right)' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

F(x)=x

If $f(x) = x$ then

$$x' = 1$$

Derivative of x^2

if $f(x) = x^2$ then $f'(x) = 2x$

Differentiate a constant value.

$$(4, 3, 2, 1)' = 0$$